

Effect of bubble dynamic behaviors on machining performance of laser-induced microjet-assisted ablation

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ABSTRACT

Laser-induced microjet-assisted ablation (LIMJAA) has been proven to be effective in fabricating high-quality surface structures. This achievement is attributed to the spontaneous generation of a continuous and stable microjet in liquid film, which prevents the recasting of ablation debris and eliminates the light-scattering effect of laser-induced cavitation bubbles. However, our experiments show that recasting occurs within the fabricated structures when laser repetition frequency exceeds a critical value in the LIMJAA process. Theoretical calculations and analyses confirm that the formation of a stable microjet requires a minimum pulse interval time exceeding the sum of the laser-induced cavitation bubble's lifetime and the microjet outflow time. This constraint of pulse interval time limits the maximum repetition frequency that LIMJAA can utilize, thereby affecting the method's overall processing efficiency. We found that reducing the liquid thickness and increasing liquid flowing velocity results in a shorter cavitation bubble lifetime and a quicker microjet outflow time, respectively, enabling shorter applicable pulse intervals. This significantly raises the usable maximum pulse repetition frequencies from approximately 40 kHz to 100 kHz. This study enhances our understanding of material removal mechanisms in LIMJAA technology and provides valuable theoretical insights for optimizing liquid-assisted laser micro-machining performance. It paves the way for advancements in high-precision and efficient laser manufacturing.

1. Introduction

Ultrafast laser beam machining has gained prominence for its ability to process materials traditionally considered challenging to machine [1–3]. However, it still faces persistent challenges, including the re-deposition of ablation debris and phase transitions [4,5]. To address these problems, liquid-assisted laser ablation techniques have been introduced. These methods can be categorized into three groups based on differences in laser-liquid coupling modes: underwater laser processing [6–8], water-guided laser processing [9–11], and waterjet-assisted laser processing [12,13]. These methods enable the cutting of a high aspect ratio and drilling of a high depth-to-diameter ratio by mitigating thermal effects and preventing debris recasting [14,15]. However, the presence of liquid environments introduces new complexities. For instance, laser-excited plasma with high temperature and high pressure can vaporize the surrounding liquid, forming cavitation bubbles [16,17]. Some results show that the suspended bubbles can reduce processing efficiency and degrade the quality of the ablated surface [18,19]. Overall, it is clear that both positive and negative effects of

liquid-assisted ablation on machining outcomes coexist, emphasizing the critical need to investigate the role of liquid in liquid-assisted ablation machining methods.

Previous studies have shown that the thickness of the liquid layer is vital in determining ablation outcomes [20]. Employing thin liquid films has proven to be an effective strategy for enhancing both machining efficiency and quality by alleviating the adverse impacts of suspended cavitation bubbles, but the mechanism is unclear [21,22]. So it can be assumed that the dynamic behavior of laser-induced cavitation bubbles plays a decisive role in liquid-assisted laser micromachining performance [23,24]. High-speed videography has enabled the tracking of the evolution of these bubbles in thin liquid films, demonstrating the significant influence of liquid film thickness on bubble dynamics [25,26]. When the liquid film thickness is slightly smaller than the bubble's maximum radius, it leads to an asymmetric evolution and collapse of the bubble [27]. This collapse, in turn, triggers a rapid ejection of liquid, a phenomenon known as a laser-induced microjet [28]. In our previous works, we have studied the formation of continuous high-speed steady microjets within a critically thin liquid film during the laser ablation

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process, and introduced a laser-induced microjet-assisted ablation (LIMJAA) technique. LIMJAA has demonstrated its capacity to enhance processing quality and achieve controllable cross-section fabrication by promptly flushing away bubbles and ablation debris using laser-induced microjets [29–31]. To further improve the processing efficiency of LIMJAA method, it is necessary to increase both the laser scanning speed and pulse repetition frequency simultaneously. However, the dynamic behaviors of laser-induced microjets and cavitation bubbles at higher pulse repetition frequencies remain inadequately understood. The instability of laser-induced microjets at high pulse repetition frequencies makes it hard to attain high processing efficiency due to the complicated coupling effects of liquid flows, laser-induced bubbles, and shock waves [32].

In this study, we systematically investigate the impact of pulse repetition frequency on the dynamic evolution of laser-induced cavitation bubbles and microjets in the LIMJAA process. Experimental investigations identify that exceeding a critical value in repetition frequency limits both laser ablation efficiency and surface quality. Subsequent theoretical analyses delve into the role of bubble lifetime and microjet outflow time in determining the applicable pulse repetition frequency for LIMJAA. Drawing from these insights, we employ a flowing liquid film with adjustable thickness and speed to modulate the bubble lifetime and microjet outflow time, respectively. Our experiments demonstrate that reducing the liquid film thickness to shorten the bubble's lifetime or increasing the liquid flow speed to accelerate the microjet, facilitates the generation of stable microjets at higher pulse repetition frequencies, ultimately enhancing the processing efficiency of LIMJAA.

2. Experimental setup and procedures

The schematic overview of the experimental setup for LIMJAA is shown in Fig. 1(a). An ultrafast laser beam wavelength λ of 520 nm and pulse duration τ_p of 300 fs is initially expanded and collimated, and subsequently focused on the sample surface using a plan achromat objective (numerical aperture NA = 0.1). The focused laser beam has a waist radius w_0 of 4.3 μm , calculated using the method proposed by Liu [33]. For workpiece manipulation, a three-dimensional motorized linear translation stage (Newport, XMS100-S) is employed. Real-time monitoring of the laser's focal point is facilitated by a coaxial CMOS camera, paired with an LED light source. Additionally, a high-speed camera (DIMAX HS1, PCO, Germany) with a maximum frame rate of 10,000 frames per second is utilized to capture the dynamic behavior of laser-induced cavitation bubbles and microjets, as elaborated in [29]. In a

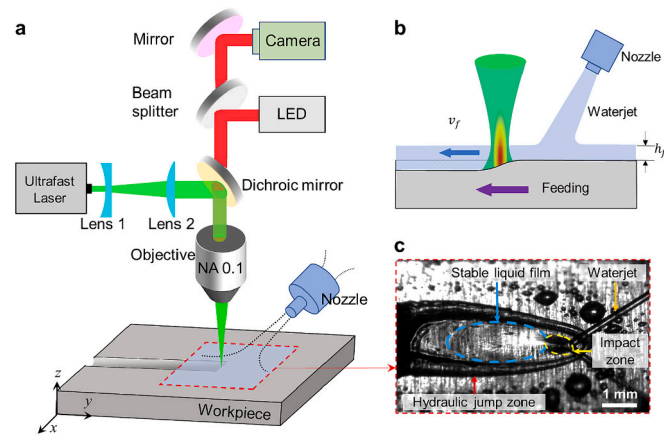


Fig. 1. Experimental setup of LIMJAA in a flowing liquid film. (a) Schematic diagram of the femtosecond laser processing system, (b) the experimental design of liquid flowing direction and workpiece feed direction, and (c) high-speed camera photo of the flowing liquid on the workpiece surface.

still liquid film, recording the time taken for a secondary bubble to travel 500 μm after the cavitation bubble collapses allows us to determine the microjet velocity as 500 μm divided by this duration.

In this study, we introduce a flowing liquid film to evaluate the influence of liquid film thickness and liquid flow speed. As illustrated in Fig. 1(b) and (c), water is ejected from an oblique nozzle and strikes the workpiece surface, forming three distinct zones: impact zone, hydraulic jump zone, and stable liquid film zone. This inconsistency of liquid film thickness in both the impact and hydraulic jump zones makes it difficult to meet the conditions necessary for generating the microjet crucial for the LIMJAA method. Therefore, the laser beam specifically targets the stable liquid film zone as shown in Fig. 1(c). Within the stable liquid film area, the fluid thickness (h_f) is approximately one-tenth of the nozzle diameter [34]. So h_f can be adjusted by altering the nozzle diameter. The liquid flowing speed (v_f) can be continuously modulated by controlling parameters such as the liquid flow rate (Q), nozzle diameter (d_n), and waterjet incident angle. In our experiments, we determined the liquid flow speed using a high-speed video camera. This was achieved by measuring the distance water waves traveled between two consecutive images and then dividing this distance by the inverse of the frame rate. The relative position of the laser beam and nozzle remains fixed while the workpiece moves along with the motorized stage.

The workpiece material used in this work is a 4H-SiC wafer with a size of 10 mm $\times$ 10 mm $\times$ 0.33 mm. The surface morphology of samples after laser ablation is examined by a scanning electron microscope (SEM, Zeiss Merlin). The dimensions and cross-sectional profiles of the fabricated microstructures are measured using a 3D laser scanning confocal microscope (LSCM, VK-X1000, Keyence). Table 1 summarizes experimental parameters for both laser processing and flowing liquid.

3. Effects of repetition frequency on the process efficiency of LIMJAA

We here experimentally explore the impact of pulse repetition frequency (f) on the groove depth of LIMJAA in a still liquid film at first. The film with a thickness of 58 μm is used in this experiment. Fig. 2 presents the groove depths and SEM images of the laser-ablated microgrooves at various feeding speeds and pulse repetition frequencies. Initially, it is observed that the groove depth exhibits an increased trend with the repetition frequency. This behavior is attributed to the logical consequence of the rise in the pulse overlap ratio. Although the ablation depth tends to increase with the laser repetition frequency, at higher frequencies (e.g., 200 kHz), the redeposition of ablation debris results in a decrease in groove depth. Specifically, for a workpiece scanning speed (v_{scan}) of 100 $\mu\text{m/s}$, the groove depth reaches its maximum of approximately 100 μm at a repetition frequency of 40 kHz, as indicated by the red dashed line in Fig. 2(a). In the low-repetition-frequency stage (≤ 40 kHz), the ablated microgrooves exhibit smooth and clean cross-sectional profiles without debris, as shown by the side view SEM images in Fig. 2(b) and (c). In contrast, in the high-repetition-frequency stage (> 40 kHz), the microgrooves' depths decrease as shown in Fig. 2(a), and recasts can be found inside the grooves, as exhibited in Fig. 2(d). Fig. 2(e)–(g) provide top views of the surface morphology of laser-ablated microgrooves at pulse repetition frequencies of 1 kHz, 10 kHz, and 100 kHz. Under the conditions of $f = 1$ kHz and $f = 10$ kHz, the microgroove

Table 1
Experimental parameters of laser processing and assistant liquid.

Parameters	Value	
	Same-/reverse-direction flow	Still liquid
Repetition frequency f (kHz)	1–200	1–200
Laser scanning speed v_{scan} ($\mu\text{m/s}$)	100, 500	100, 500
Pulse energy E_p (μJ)	0–20	0–20
Liquid film flowing speed v_f (m/s)	5, 8, 11, 14, 17, 20	0
Liquid film thickness h_f (μm)	14, 21, 26, 34, 42	58, 75

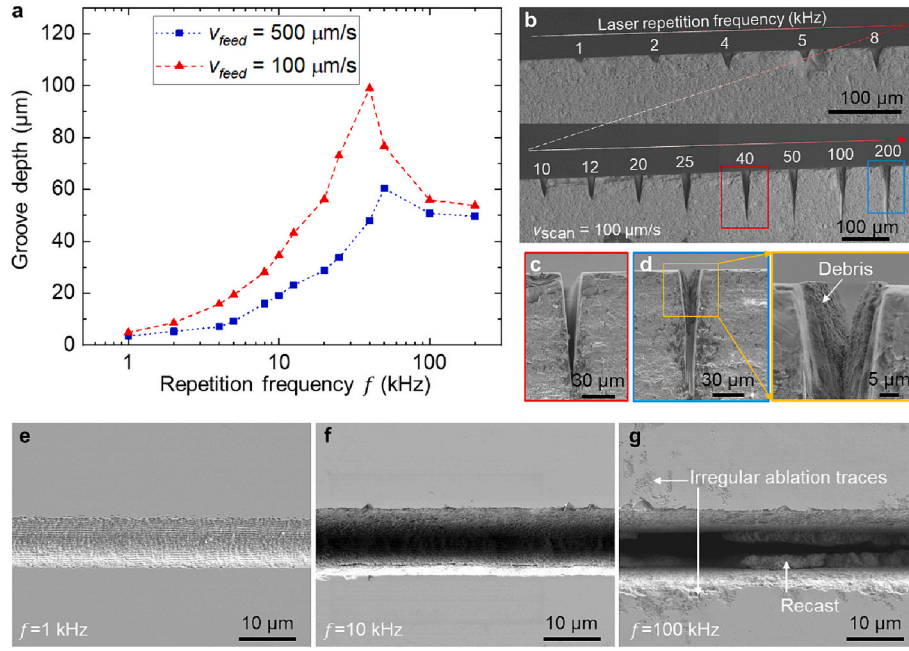


Fig. 2. The ablation performance of LIMJAA in a still liquid film with the increase of pulse repetition frequency. (a) Groove depths, (b) cross-sectional profiles, (c), (d) enlarged side views, and (e), (f), (g) surface morphology of microgrooves ablated by LIMJAA in a still liquid film at varying laser repetition frequencies. Pulse energies are set to 12 μJ .

surfaces are smooth and free of recast and redeposition, as shown in Fig. 2(e) and (f). However, when the pulse repetition frequency is increased to 100 kHz, recast and redeposition become noticeable within the ablated microgrooves, as illustrated in Fig. 2(g). Additionally, some unexpected ablation traces are visible around the grooves, possibly resulting from laser light refracted by bubbles and shock waves induced by bubble collapse.

To gain deeper insights into the influence of varying repetition frequencies on processing quality, we used a high-speed camera to capture the evolution of laser-induced microjets. In these experiments, three types of bubbles are present. Firstly, laser-induced cavitation bubbles, which have a lifetime close to 10 μs , are challenging to observe due to their rapid formation and collapse, which exceeds our camera's capture speed of 100 $\mu\text{s}/\text{frame}$. Secondly, the secondary bubbles formed by the collapse of cavitation bubbles are depicted in Fig. 3(a) and (b). Thirdly, large bubbles result from merging an un-collapsed bubble with the subsequent cavitation bubble, as illustrated in Fig. 3(c). Among these, the directional movement of secondary bubbles can be used to characterize the flow direction and speed of microjets. The obvious bubble movement observed in Fig. 3(a) and (b) indicate that stable microjets are

formed at 1 kHz and 10 kHz, while the bubble coalescence seen in Fig. 3 (c) suggests the interaction between two pulse-induced cavitation bubbles. This bubble cluster impedes the formation of laser-induced microjets and induces a light scattering effect, subsequently deteriorating the surface quality and diminishing the ablation efficiency, as shown in Fig. 2(g). From these observations, it can be concluded that repetition frequency surpassing a certain threshold is detrimental to the stability of laser processing.

For a stable ablation process in LIMJAA, it is imperative to remove the ablation products and bubbles from the ablation zone effectively through the laser-induced microjet. To meet this requirement, the following conditions must be satisfied: 1) the cavitation bubble should completely collapse before the next pulse arrives, and 2) the collapse-induced secondary bubbles are flushed away by the laser-induced microjet before the next pulse arrives. These two conditions can be ensured when the laser pulse interval time ($\Delta t = 1/f$) satisfies the following relationship,

$$\Delta t > T_B + T_{\text{out}} \quad (1)$$

where T_B is bubble lifetime, and T_{out} corresponds to microjet outflow time which denotes the duration for the microjet to flow out from the bubble effect zone. The bubble effect zone is defined as the maximum area a cavitation bubble can reach, making the bubble effect zone width equal to the maximum bubble diameter.

The exploration of cavitation bubble evolution in an immersed liquid was first undertaken by Plesset et al. [35] as a means to investigate cavitation damage in pumps, turbines, and propellers. Subsequently, the dynamic evolution of a single cavitation bubble's expansion and contraction was described by the Rayleigh-Plesset model [36]. In the collapse stage of a cavitation bubble, Rayleigh's collapse time T_C can be deduced from the maximum bubble radius R_{max} as follows [37],

$$T_C = 0.915 R_{\text{max}} \sqrt{\frac{\rho}{P_{\infty} - P_v}} \quad (2)$$

where ρ is liquid density ($\sim 980 \text{ kg/m}^3$ for pure water), P_{∞} represents the surrounding liquid pressure ($\sim 0.101 \text{ MPa}$), and P_v is the saturated vapor pressure of the bubble with maximum radius, which is small

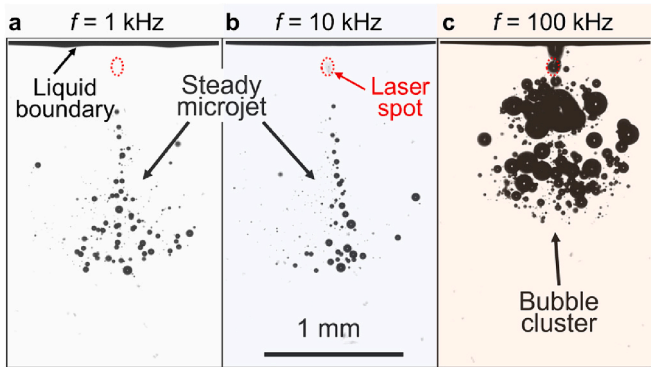


Fig. 3. Shadowgraphs of laser-induced cavitation bubbles and microjets after 50-ms laser irradiation with various repetition frequencies. (a) 1 kHz, (b) 10 kHz, and (c) 100 kHz. The pulse energy is 12 μJ .

enough to be ignored compared with P_∞ .

In LIMJAA, the laser-induced cavitation bubble is generated near a free surface boundary rather than in infinite liquid. The presence of a workpiece can sometimes affect the collapse time of bubbles. But when the boundary distance is less than or equal to the maximum radius of the bubble, the collapse time of the bubble no longer increases [38]. Laser focus on the workpiece in these experiments, so the effect of the workpiece was not factored into our bubble lifetime calculations. The presence of the free boundary alters the liquid pressure around the cavitation bubble, affecting the kinetic energy of the cavitation bubble and, consequently, its collapse time. This leads to a shortened bubble collapse time T_{CF} , which is determined by the shortening factor κ_s , as verified by Gregorić [38],

$$\kappa_s = \frac{T_{CF}}{T_C} = 1 - \frac{0.102}{\gamma_F} \quad (3)$$

where γ_F is a dimensionless parameter that denotes the relative distance from the bubble center to the liquid surface,

$$\gamma_F = h/R_{\max} \quad (4)$$

where h is the distance from the bubble center to the liquid surface. In this work, the centers of laser-induced cavitation bubbles are always located on the workpiece surface, thus h is equal to the thickness of the liquid film h_f .

A cavitation bubble will experience expansion and collapse processes successively, thus the lifetime of a bubble T_B is twice the collapse time T_C , i.e., $T_B = 2T_C$. Therefore, the shortened lifetime T_B of a laser-induced bubble near the free boundary can be expressed as,

$$T_B = 2\kappa_s T_C = 1.83 \left(1 - \frac{0.102}{h_f/R_{\max}} \right) R_{\max} \sqrt{\frac{\rho}{P_\infty - P_v}} \quad (5)$$

The pulse energy of the laser beam affects the bubble's lifetime T_B since it largely determines the maximum bubble radius $R_{\max}$. To determine this, we conducted experiments to measure the maximum bubble radius $R_{\max}$ at different laser pulse energies E_p using the high-speed camera with a 16× microscope objective and the results are shown in Fig. 4(a). The cavitation bubble, having an elliptical shape, can be equivalently represented by a sphere with the same volume. The radius of this sphere is considered the maximum bubble radius $R_{\max}$:

$$R_{\max} = \sqrt[3]{R_x^2 R_z} \quad (6)$$

where R_x defines the horizontal radius of the cavitation bubble and R_z defines the vertical radius of the cavitation bubble. Based on the measured maximum bubble radius and Eq. (5), the theoretical bubble lifetime is calculated, as shown in Fig. 4(b). It is evident that the calculated bubble lifetime increases with the liquid film thickness h_f . Consequently, we propose a strategy to artificially manipulate the

bubble lifetime by adjusting the film thickness. Fig. 4(b) also displays various curves depicting bubble lifetimes at different laser pulse energies (4–20 μJ).

The second factor influencing the stability of LIMJAA is the microjet outflow time (T_{out}). It requires enough time for collapse-induced secondary bubbles to be flushed away by the laser-induced microjet before the next pulse arrives. T_{out} can be estimated as

$$T_{out} = D_{\max}/v_l \quad (7)$$

where $D_{\max}$ is the diameter of the maximum cavitation bubble dimension ($D_{\max} = 2R_{\max}$) and v_l is the local liquid flowing speed in the laser ablation zone. The local liquid flowing speed v_l equals to a microjet speed v_{mj} if a still liquid film is used. Otherwise, if utilizing a flowing liquid film that aligns with the workpiece's feeding direction, the local liquid flowing speed v_l is the sum of the initial liquid film speed v_f and microjet speed v_{mj} . The same-direction flowing liquid film helps decrease the outflow time, while the reverse-direction flowing liquid film increases the outflow time.

If Δt satisfies the condition of Eq. (1), it implies that the laser-induced microjet can effectively remove ablation debris and bubbles before the next laser pulse arrives, ensuring stable LIMJAA processing. Conversely, if the repetition frequency f exceeds a critical value and the corresponding interval time Δt ($=1/f$) becomes too short, falling below the sum of the bubble lifetime and microjet outflow time ($\Delta t < T_B + T_{out}$), the processing becomes unstable. Therefore, a theoretical maximum applicable repetition frequency $f_{\max}$ that ensures a continuous laser-induced microjet and stable machining process can be defined as

$$f_{\max} = 1/(T_B + T_{out}) \quad (8)$$

The bubbles' lifetime T_B varying with laser pulse energy and liquid thickness is estimated through Eqs. (2)–(5), as shown in Fig. 4(b). The microjet outflow time T_{out} can be acquired from Eq. (7). Thus, the $f_{\max}$ can be calculated. For example, when the laser pulse energy is 12 μJ and the liquid film thickness is 75 μm, the measured maximum bubble radius $R_{\max}$ and dimensionless distance γ_F are 75 μm and 1, respectively. Using the high-speed camera as described in our previous work [29], the measured microjet speed v_{mj} is 5.5 m/s. The corresponding bubble lifetime and microjet outflow time are $T_B = 12.2$ μs and $T_{out} = 27.3$ μs. Thus, according to Eq. (7), the calculated theoretical maximum applicable repetition frequency $f_{\max}$ is 25.3 kHz. In theory, when the laser pulse repetition frequency exceeds this value, the laser-induced microjet becomes discontinuous and unstable. The theoretical calculation results agree with the experimental results in Fig. 2 and Fig. 3.

From Eq. (8), it becomes apparent that increasing the maximum applicable repetition frequency can be achieved by minimizing either the bubble lifetime T_B or the microjet outflow time T_{out} . The bubble lifetime is predominantly determined by the thickness of the liquid film, while the microjet T_{out} can be diminished by accelerating the speed of the laser-induced microjet. Therefore, this study recommends the

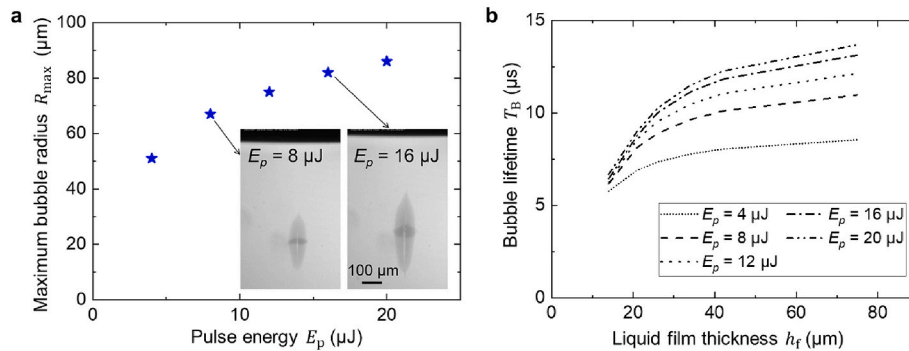


Fig. 4. Maximum bubble radius test and bubble lifetime calculation. (a) Measured maximum bubble radius, and (b) calculated bubble lifetime varies with liquid film thickness at different pulse energies.

utilization of a flowing liquid film with adjustable thickness to shorten the bubble lifetime T_B . Simultaneously, the flowing liquid film enhances the local liquid flowing speed and shortens the microjet outflow time T_{out} , enabling the attainment of higher applicable repetition frequencies and improving stability in laser processing.

4. Improvement of processing efficiency of LIMJAA

As depicted in Section 3, both the flowing speed and thickness of liquid film significantly influence the maximum applicable repetition frequency f_{max} in LIMJAA. To improve the f_{max} in the experiment, we use an oblique incident water jet that strikes the workpiece surface, resulting in the formation of a flowing liquid film with an adjustable thickness.

4.1. Effect of liquid film thickness on processing efficiency of LIMJAA

To investigate the impact of liquid film thickness, we set a constant liquid flowing speed ($v_f = 10$ m/s) and variable liquid film thickness by adjusting the nozzle diameter d_n and waterjet flow rate Q . Fig. 5(a) illustrates the variation in ablated microgroove depth with pulse repetition frequency for different liquid film thicknesses. Each data point is the average value of five microgrooves ablated under the same conditions. As the repetition frequency increases, the microgroove depth initially exhibits an upward trend before decreasing.

Fig. 5(b) presents a summary of the maximum applicable repetition frequency F_{max} derived from Fig. 5(a), along with the calculated theoretical maximum repetition frequency f_{max} for various liquid film thicknesses. When the thickness of the liquid film is reduced from 75 μm to 21 μm , F_{max} increases from 40 kHz to 55 kHz. This increase in thinner liquid films is attributed to the shortened bubble lifetime, which allows for shorter pulse interval time and higher repetition frequency for stable

processing. However, a notable decline in F_{max} occurs at a much thinner liquid thickness of 14 μm , even lower than that in a still-liquid film. As shown in Fig. 5(c) and (d), the movement speed of the bubbles is used to characterize the microjet speed. It can be calculated that for a liquid film thickness of 14 μm , the maximum speed of the microjet is close to 0.015 m/s. In contrast, for a liquid film thickness of 26 μm , the bubble maintains a considerable speed, approximately 0.075 m/s, even after moving 300 μm . This observation demonstrates that the thickness of the liquid film significantly influences microjet formation, which in turn affects the removal of ablation products and ultimately the depth of the grooves. It can be inferred that the bubble breaks through the film's upper interface and collapses before it expands to its maximum volume, so the laser-induced microjet cannot be generated, and the process of LIMJAA is disturbed.

The experimental findings and theoretical analyses collectively demonstrate that the LIMJAA approach in a fluid film can moderately increase the maximum applicable repetition frequency by reducing the liquid film thickness. However, it is essential to maintain a sufficient liquid film thickness to prevent cavitation bubbles from penetrating the liquid film before collapsing, ensuring the continuous generation of laser-induced microjets.

4.2. Effect of liquid film flowing speed on processing efficiency of LIMJAA

Another factor influencing the maximum applicable repetition frequency F_{max} is the liquid film flowing speed v_f , which changes the outflow time T_{out} of the ablation products and bubbles from the ablation zone. In this study, we experimentally investigate the effect of the liquid film flow speed on the maximum applicable repetition frequency, as shown in Fig. 6. The dotted and dashed lines represent the calculated theoretical maximum applicable repetition frequency in liquid films

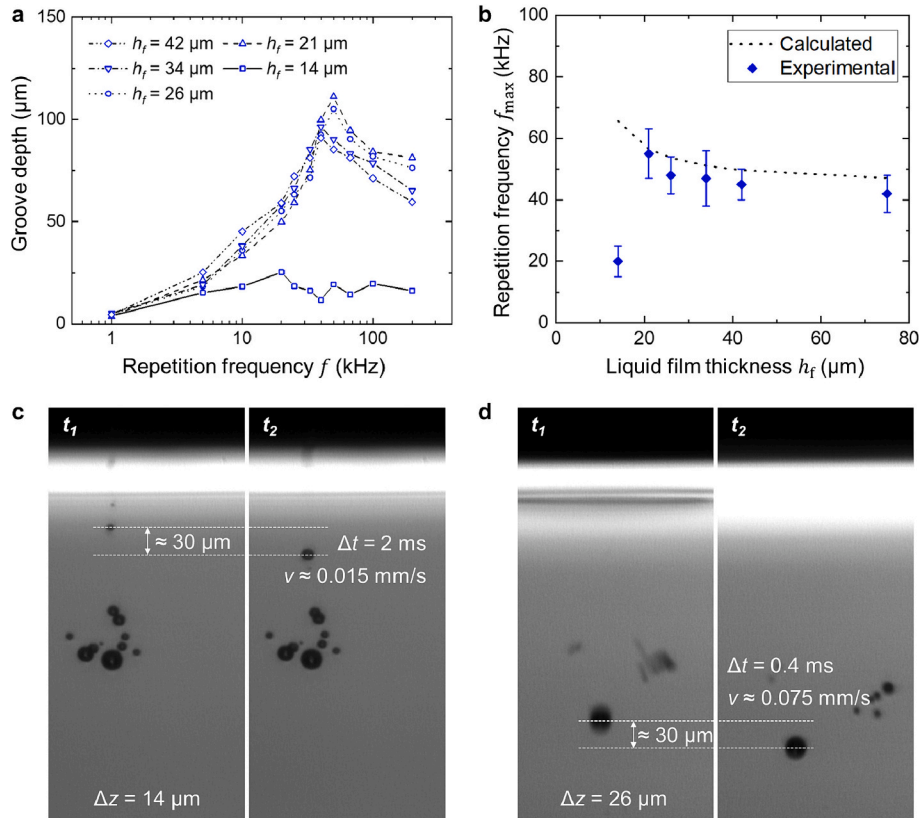


Fig. 5. Influence of flowing liquid film thickness on laser ablation performance. (a) Test groove depth varying with laser repetition frequency and liquid film thickness, (b) maximum applicable pulse repetition frequency changing with liquid film thickness. (c) and (d) are photographs of laser-induced microjet when laser focusing at 14 μm and 26 μm below the liquid interface, respectively. The laser pulse energy is 12 μJ and the workpiece feed speed is 100 $\mu\text{m/s}$.

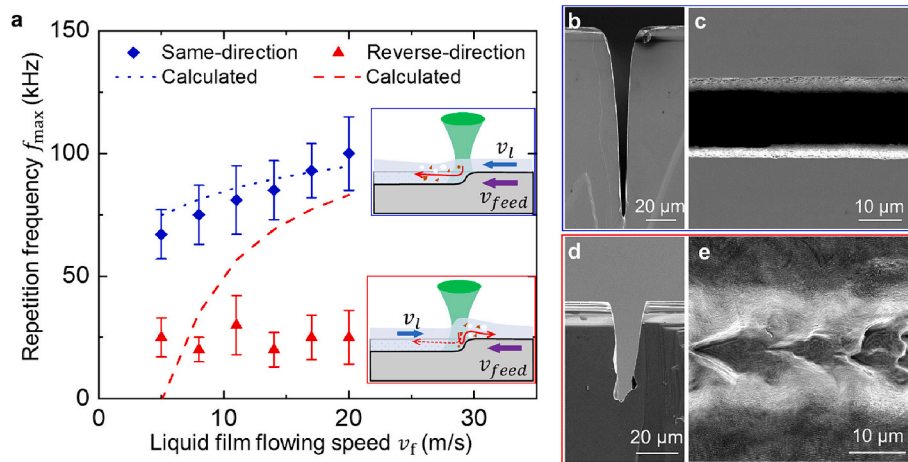


Fig. 6. Influence of liquid film flowing speed on laser ablation performance. (a) Effect of liquid film flowing speed on the maximum applicable repetition frequency with different flowing directions. (b) Side view and (c) top viewed SEM images of microgroove ablated in same-direction fluid; (d) side view and (e) top view SEM images of microgroove ablated in the reverse-direction fluid. The repetition frequency and liquid film flowing speed are kept the same (100 kHz and 20 m/s, respectively). The laser pulse energy is 12 μJ , and the workpiece feed speed is 100 $\mu\text{m/s}$. The liquid film thickness is controlled to be 21 μm .

with the same and reversed flowing direction, respectively.

When the liquid flows in the same direction as the workpiece feeding direction, as shown in Fig. 6(a), $F_{\max}$ increases with higher liquid flow speed v_f . The insert schematic image in the upper right of Fig. 6(a) illustrates that when the fluid flows in the same direction, it accelerates the laser-induced microjet. A faster laser-induced microjet enhances the removal of debris and secondary bubbles, consequently reducing the outflow time T_{out} . This reduction in T_{out} contributes to an improved maximum applicable repetition frequency $F_{\max}$ according to Eq. (8). Thus, a higher repetition frequency can be employed for stable laser processing in a faster-flowing liquid film. For a flowing liquid film with $v_f = 20$ m/s, a 100-kHz-repetition-frequency laser beam can conduct stable line-scanning ablation. A 125- μm -depth deep microgroove with a clean surface is fabricated as shown in Fig. 6(b) and (c).

Conversely, $F_{\max}$ significantly decreases when the liquid flows in the opposite direction to the workpiece feeding direction, as shown in Fig. 6(a). In this condition, the initial liquid flow decelerates the laser-induced microjet, which slows down the laser-induced microjet, as shown by the inserted schematic image in the lower right of Fig. 6(a). This reverse-direction flow causes a hydraulic jump when it encounters the ablation front, which results in the disturbance of the laser beam. The theoretical calculation does not account for the hydraulic jump effect, leading to a much higher theoretical maximum repetition frequency than observed in the experiments. Consequently, a longer pulse interval is required to stabilize the liquid and remove debris and bubbles. The applicable repetition frequency varies irregularly from 20 to 30 kHz as the liquid film speed ranges from 5 to 20 m/s. When utilizing a laser beam with a repetition frequency of 100 kHz and a flowing liquid film with $v_f = 20$ m/s (keeping other parameters consistent with the same-direction flows), with a shallower ablation depth compared to the same feeding direction condition, as evident in Fig. 6(d) and (e). Therefore, it is essential for the liquid flow direction to always align with the sample feed direction to achieve optimal processing quality and efficiency when utilizing the LIMJAA method in processing with complex scanning paths.

5. Conclusion

This study systematically investigates the dynamic evolution characteristics of laser-induced cavitation bubbles and microjets and their influence on the performance of laser-induced microjet-assisted ablation (LIMJAA). By employing a flowing liquid film with adjustable film thickness and flowing speed, the applicable laser pulse repetition

frequency has been largely improved. The key conclusions drawn from this study are as follows:

- (1) In the LIMJAA process, if laser pulse repetition frequency exceeds the critical value, the ablation efficiency reduces, and the surface quality deteriorates. High-speed camera results demonstrate that the accumulation and growth of bubbles obstruct the subsequent laser beam transmission.
- (2) Theoretical calculations reveal that as the pulse repetition frequency surpasses the critical value, the pulse interval becomes too short to effectively remove bubbles and debris from the ablation zone, thus suppressing the ablation efficiency. The maximum applicable repetition frequency is influenced by factors including cavitation bubble lifetime and microjet outflow time.
- (3) A thinner liquid film results in a shorter bubble lifetime, enabling a higher applicable pulse repetition frequency while maintaining stable laser processing performance. However, there is a minimum allowable liquid thickness, as a sufficient liquid film thickness is required to prevent bubbles from breaching the liquid boundary before reaching their maximum size.
- (4) Liquid flow along the same direction of workpiece feeding is advantageous for promptly removing bubbles from the ablation zone, allowing for a reduced pulse interval. Consequently, the applicable pulse repetition frequency is enhanced from approximately 40 kHz to 100 kHz, leading to improved processing efficiency.

The findings presented in this study offer valuable insights into understanding the evolution characteristics of bubbles and microjets in liquid-assisted laser micromachining. Furthermore, the mechanisms and approaches presented pave the way for achieving high-performance liquid-assisted laser ablation in the machining of hard-to-machine materials.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmapro.2024.04.041>.

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